K-MEANS CLUSTERING ALGORITHM

Name: _____G VYSHNAVI_____ **REG No.: 192525260**_____

Subject: _____DWDM_____ **Date:** __30/10/26_____

K-Means Clustering Algorithm – Numerical Problems

Instructions: Solve the following problems using the K-Means Clustering Algorithm. Show all iterations, distance calculations, cluster assignments, and updated centroids.

Problem 1 – K = 3

Consider the following dataset:

$$D=\{2,4,6,8,10,12,14,16,18\}$$

Apply the K-Means algorithm with K = 3.

Initial centroids:

- C1 = 2
- C2 = 8
- C3 = 14

Find:Distance of each data point from all centroids.Cluster assignment.New centroid values.Final clusters after convergence.

Problem 2 – K = 4

Consider the following dataset representing the daily sales of a company (in hundreds):

$$D=\{5,10,15,20,25,30,35,40,45,50\}$$

Apply the K-Means algorithm with K = 4.

Initial centroids:

- C1 = 5
- C2 = 15
- C3 = 30
- C4 = 45

Find:Euclidean distance between data points and centroids.Assign data points to their nearest clusters.Calculate updated centroids.Repeat until convergence and identify the final clusters.

Problem 3 – K = 5

Consider the following dataset representing the number of customers visiting a shopping mall per hour:

$$D=\{3,6,9,15,18,24,27,33,36,42\}$$

Apply the K-Means algorithm with K = 5.

Initial centroids:

- C1 = 3
- C2 = 9
- C3 = 18
- C4 = 27
- C5 = 36

Find:Calculate the distance of every data point from all five centroids.Assign each data point to the nearest cluster.Calculate the new centroid for each cluster.Repeat the iterations until convergence.Write the final clusters and centroids.

Problem 1 – K = 3

Given data: D = {2, 4, 6, 8, 10, 12, 14, 16, 18}

Initial centroids: C1 = 2, C2 = 8, C3 = 14

Distance in one dimension: $d(x, Cj) = |x - Cj|$

Updated centroid: Cj = (sum of all points assigned to cluster j) / (number of points in that cluster).

Iteration 1

Centroids used: C1 = 2, C2 = 8, C3 = 14

Data point	d to C1	d to C2	d to C3	Assigned cluster
2	0	6	12	C1
4	2	4	10	C1
6	4	2	8	C2
8	6	0	6	C2
10	8	2	4	C2
12	10	4	2	C3
14	12	6	0	C3
16	14	8	2	C3
18	16	10	4	C3

Cluster assignment: C1: {2, 4} C2: {6, 8, 10} C3: {12, 14, 16, 18}

Centroid update

C1 = (2 + 4) / 2 = 3

C2 = (6 + 8 + 10) / 3 = 8

C3 = (12 + 14 + 16 + 18) / 4 = 15

Updated centroids: C1 = 3, C2 = 8, C3 = 15

The centroids changed, so another iteration is required.

Iteration 2

Centroids used: C1 = 3, C2 = 8, C3 = 15

Data point	d to C1	d to C2	d to C3	Assigned cluster
2	1	6	13	C1
4	1	4	11	C1
6	3	2	9	C2
8	5	0	7	C2
10	7	2	5	C2
12	9	4	3	C3
14	11	6	1	C3
16	13	8	1	C3
18	15	10	3	C3

Cluster assignment: C1: {2, 4} C2: {6, 8, 10} C3: {12, 14, 16, 18}

Centroid update
C1 = (2 + 4) / 2 = 3

C2 = (6 + 8 + 10) / 3 = 8

C3 = (12 + 14 + 16 + 18) / 4 = 15

Updated centroids: C1 = 3, C2 = 8, C3 = 15

The next iteration gives the same assignments and centroids. Therefore, the algorithm has converged.

Final Answer

Cluster 1 = {2, 4}; Final centroid C1 = 3.

Cluster 2 = {6, 8, 10}; Final centroid C2 = 8.

Cluster 3 = {12, 14, 16, 18}; Final centroid C3 = 15.

Hence, the above clusters and centroids are the final result after convergence.

Problem 2 – K = 4

Given data (daily sales in hundreds): D = {5, 10, 15, 20, 25, 30, 35, 40, 45, 50}

Initial centroids: C1 = 5, C2 = 15, C3 = 30, C4 = 45

Distance in one dimension: $d(x, Cj) = |x - Cj|$

Updated centroid: Cj = (sum of all points assigned to cluster j) / (number of points in that cluster).

Iteration 1

Centroids used: C1 = 5, C2 = 15, C3 = 30, C4 = 45

Data point	d to C1	d to C2	d to C3	d to C4	Assigned cluster
5	0	10	25	40	C1
10	5	5	20	35	C1
15	10	0	15	30	C2
20	15	5	10	25	C2
25	20	10	5	20	C3
30	25	15	0	15	C3
35	30	20	5	10	C3
40	35	25	10	5	C4
45	40	30	15	0	C4
50	45	35	20	5	C4

Cluster assignment: C1: {5, 10} C2: {15, 20} C3: {25, 30, 35} C4: {40, 45, 50}

Centroid update

C1 = (5 + 10) / 2 = 7.5

C2 = (15 + 20) / 2 = 17.5

C3 = (25 + 30 + 35) / 3 = 30

C4 = (40 + 45 + 50) / 3 = 45

Updated centroids: C1 = 7.5, C2 = 17.5, C3 = 30, C4 = 45

The centroids changed, so another iteration is required.

Iteration 2

Centroids used: C1 = 7.5, C2 = 17.5, C3 = 30, C4 = 45

Data point	d to C1	d to C2	d to C3	d to C4	Assigned cluster
5	2.5	12.5	25	40	C1
10	2.5	7.5	20	35	C1
15	7.5	2.5	15	30	C2
20	12.5	2.5	10	25	C2
25	17.5	7.5	5	20	C3
30	22.5	12.5	0	15	C3
35	27.5	17.5	5	10	C3
40	32.5	22.5	10	5	C4
45	37.5	27.5	15	0	C4
50	42.5	32.5	20	5	C4

Cluster assignment: C1: {5, 10} C2: {15, 20} C3: {25, 30, 35} C4: {40, 45, 50}

Centroid update

C1 = (5 + 10) / 2 = 7.5

C2 = (15 + 20) / 2 = 17.5

C3 = (25 + 30 + 35) / 3 = 30

C4 = (40 + 45 + 50) / 3 = 45

Updated centroids: C1 = 7.5, C2 = 17.5, C3 = 30, C4 = 45

The next iteration gives the same assignments and centroids. Therefore, the algorithm has converged.

Final Answer

Cluster 1 = {5, 10}; Final centroid C1 = 7.5.

Cluster 2 = {15, 20}; Final centroid C2 = 17.5.

Cluster 3 = {25, 30, 35}; Final centroid C3 = 30.

Cluster 4 = {40, 45, 50}; Final centroid C4 = 45.

Hence, the above clusters and centroids are the final result after convergence.

Problem 3 – K = 5

Given data (customers per hour): D = {3, 6, 9, 15, 18, 24, 27, 33, 36, 42}

Initial centroids: C1 = 3, C2 = 9, C3 = 18, C4 = 27, C5 = 36

Distance in one dimension: d(x, Cj) = |x − Cj|

Updated centroid: Cj = (sum of all points assigned to cluster j) / (number of points in that cluster).

Iteration 1

Centroids used: C1 = 3, C2 = 9, C3 = 18, C4 = 27, C5 = 36

Data point	d to C1	d to C2	d to C3	d to C4	d to C5	Assigned cluster
3	0	6	15	24	33	C1
6	3	3	12	21	30	C1
9	6	0	9	18	27	C2
15	12	6	3	12	21	C3
18	15	9	0	9	18	C3
24	21	15	6	3	12	C4
27	24	18	9	0	9	C4
33	30	24	15	6	3	C5
36	33	27	18	9	0	C5
42	39	33	24	15	6	C5

Cluster assignment: C1: {3, 6} C2: {9} C3: {15, 18} C4: {24, 27} C5: {33, 36, 42}

Centroid update

C1 = (3 + 6) / 2 = 4.5

C2 = (9) / 1 = 9

C3 = (15 + 18) / 2 = 16.5

C4 = (24 + 27) / 2 = 25.5

C5 = (33 + 36 + 42) / 3 = 37

Updated centroids: C1 = 4.5, C2 = 9, C3 = 16.5, C4 = 25.5, C5 = 37

The centroids changed, so another iteration is required.

Iteration 2

Centroids used: C1 = 4.5, C2 = 9, C3 = 16.5, C4 = 25.5, C5 = 37

Data point	d to C1	d to C2	d to C3	d to C4	d to C5	Assigned cluster
3	1.5	6	13.5	22.5	34	C1
6	1.5	3	10.5	19.5	31	C1
9	4.5	0	7.5	16.5	28	C2
15	10.5	6	1.5	10.5	22	C3
18	13.5	9	1.5	7.5	19	C3
24	19.5	15	7.5	1.5	13	C4
27	22.5	18	10.5	1.5	10	C4
33	28.5	24	16.5	7.5	4	C5
36	31.5	27	19.5	10.5	1	C5
42	37.5	33	25.5	16.5	5	C5

Cluster assignment: C1: {3, 6} C2: {9} C3: {15, 18} C4: {24, 27} C5: {33, 36, 42}

Centroid update

C1 = (3 + 6) / 2 = 4.5

C2 = (9) / 1 = 9

C3 = (15 + 18) / 2 = 16.5

C4 = (24 + 27) / 2 = 25.5

C5 = (33 + 36 + 42) / 3 = 37

Updated centroids: C1 = 4.5, C2 = 9, C3 = 16.5, C4 = 25.5, C5 = 37

The next iteration gives the same assignments and centroids. Therefore, the algorithm has converged.

Final Answer

Cluster 1 = {3, 6}; Final centroid C1 = 4.5.

Cluster 2 = {9}; Final centroid C2 = 9.

Cluster 3 = {15, 18}; Final centroid C3 = 16.5.

Cluster 4 = {24, 27}; Final centroid C4 = 25.5.

Cluster 5 = {33, 36, 42}; Final centroid C5 = 37.

Hence, the above clusters and centroids are the final res